

Exam 3 Questions (6.A – 19.C)
CHEM 1310 | Spring 2026

QUESTION 1 (5 points)

One mole of an ideal gas with heat capacity $0.205 \text{ L}\cdot\text{atm}/^\circ\text{C}$ is heated at a constant pressure of 0.980 atm from 20.0°C to 95.0°C . The gas expands from a volume of 0.500 L to 4.800 L . What is the change in internal energy (ΔU) of the gas for this process?

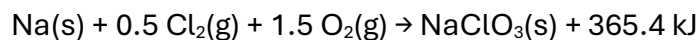
- A. ☐ **+11.2 L·atm**
 - B. ☐ +15.4 L·atm
 - C. ☐ -11.2 L·atm
 - D. ☐ +19.6 L·atm
 - E. ☐ -1.13 L·atm
-

One mole of an ideal gas with heat capacity $0.205 \text{ L}\cdot\text{atm}/^\circ\text{C}$ is heated at a constant pressure of 0.980 atm from 20.0°C to 45.0°C . The gas expands from a volume of 0.500 L to 4.600 L . What is the change in internal energy (ΔU) of the gas for this process?

- A. ☐ **+1.11 L·atm**
 - B. ☐ +5.13 L·atm
 - C. ☐ +4.02 L·atm
 - D. ☐ +9.15 L·atm
 - E. ☐ -2.85 L·atm
-

QUESTION 2 (5 points)

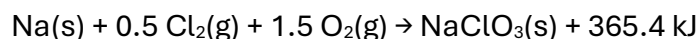
Formation of solid sodium chlorate from its elements is an exothermic process with ΔH° equal to -365.4 kJ/mol .



In one preparation of NaClO_3 using this reaction, 80.2 kJ of heat were evolved. How many moles of NaClO_3 were produced? Enter your response in moles (mol) to the nearest 0.01 mol .

0.22 mol

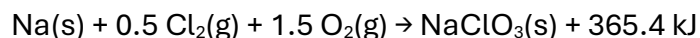
Formation of solid sodium chlorate from its elements is an exothermic process with ΔH° equal to -365.4 kJ/mol .



In one preparation of NaClO_3 using this reaction, 412 kJ of heat were evolved. How many moles of NaClO_3 were produced? Enter your response in moles (mol) to the nearest 0.01 mol .

1.13 mol

Formation of solid sodium chlorate from its elements is an exothermic process with ΔH° equal to -365.4 kJ/mol .



In one preparation of NaClO_3 using this reaction, 845 kJ of heat were evolved. How many moles of NaClO_3 were produced? Enter your response in moles (mol) to the nearest 0.01 mol .

2.31 mol

QUESTION 3 (5 points)

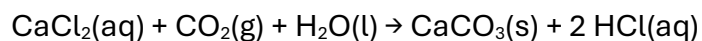
Use the standard enthalpies of formation given to determine the standard enthalpy of reaction for the process below.



Substance	ΔH°_f (kJ/mol)
$\text{CaCO}_3(\text{s})$	-1207
$\text{HCl}(\text{aq})$	-167
$\text{CaCl}_2(\text{aq})$	-877
$\text{CO}_2(\text{g})$	-394
$\text{H}_2\text{O}(\text{l})$	-286

- A. ☐ -16 kJ
B. ☐ -350 kJ
C. ☐ -270 kJ
D. ☐ -1557 kJ
E. ☐ +1541 kJ
-

Use the standard enthalpies of formation given to determine the standard enthalpy of reaction for the process below.

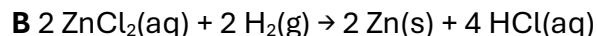
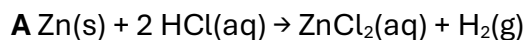


Substance	ΔH°_f (kJ/mol)
$\text{CaCO}_3(\text{s})$	-1207
$\text{HCl}(\text{aq})$	-167
$\text{CaCl}_2(\text{aq})$	-877
$\text{CO}_2(\text{g})$	-394
$\text{H}_2\text{O}(\text{l})$	-286

- A. ☐ +16 kJ
B. ☐ +350 kJ
C. ☐ +270 kJ
D. ☐ +1557 kJ
E. ☐ -1541 kJ

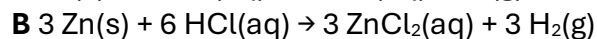
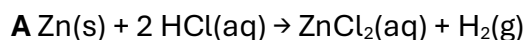
QUESTION 4 (5 points)

Consider reactions **A** and **B** below. How is the enthalpy of reaction **B** related mathematically to the enthalpy of reaction **A**?



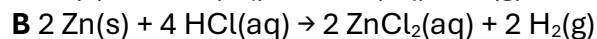
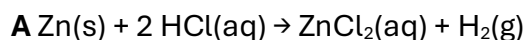
- A. ☐ $\Delta H^\circ(\text{B}) = -2\Delta H^\circ(\text{A})$
 - B. ☐ $\Delta H^\circ(\text{B}) = -\Delta H^\circ(\text{A})$
 - C. ☐ $\Delta H^\circ(\text{B}) = +2\Delta H^\circ(\text{A})$
 - D. ☐ $\Delta H^\circ(\text{B}) = +3\Delta H^\circ(\text{A})$
 - E. ☐ The enthalpies are not related at all.
-

Consider reactions **A** and **B** below. How is the enthalpy of reaction **B** related mathematically to the enthalpy of reaction **A**?



- A. ☐ $\Delta H^\circ(\text{B}) = +3\Delta H^\circ(\text{A})$
 - B. ☐ $\Delta H^\circ(\text{B}) = -2\Delta H^\circ(\text{A})$
 - C. ☐ $\Delta H^\circ(\text{B}) = -\Delta H^\circ(\text{A})$
 - D. ☐ $\Delta H^\circ(\text{B}) = +2\Delta H^\circ(\text{A})$
 - E. ☐ The enthalpies are not related at all.
-

Consider reactions **A** and **B** below. How is the enthalpy of reaction **B** related mathematically to the enthalpy of reaction **A**?



- A. ☐ $\Delta H^\circ(\text{B}) = +2\Delta H^\circ(\text{A})$
- B. ☐ $\Delta H^\circ(\text{B}) = +3\Delta H^\circ(\text{A})$
- C. ☐ $\Delta H^\circ(\text{B}) = -2\Delta H^\circ(\text{A})$
- D. ☐ $\Delta H^\circ(\text{B}) = -\Delta H^\circ(\text{A})$
- E. ☐ The enthalpies are not related at all.

QUESTION 5 (6 points)

Three metals **A – C** with equal mass are placed on a hot plate, which provides a constant amount of heat to each metal. Initial and final temperatures of each metal are listed below. Rank the metals from smallest (1) to largest (3) specific heat.

Metal	Initial Temperature (°C)	Final Temperature (°C)
A	20.0	38.2
B	20.0	47.1
C	20.0	62.3

1 smallest specific heat **C**
2 **B**
3 largest specific heat **A**

Three metals **A – C** with equal mass are placed on a hot plate, which provides a constant amount of heat to each metal. Initial and final temperatures of each metal are listed below. Rank the metals from smallest (1) to largest (3) specific heat.

Metal	Initial Temperature (°C)	Final Temperature (°C)
A	20.0	48.2
B	20.0	37.1
C	20.0	62.3

1 smallest specific heat **C**
2 **A**
3 largest specific heat **B**

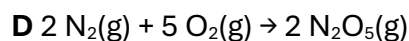
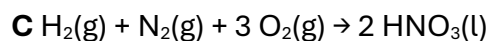
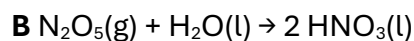
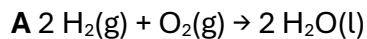
Three metals **A – C** with equal mass are placed on a hot plate, which provides a constant amount of heat to each metal. Initial and final temperatures of each metal are listed below. Rank the metals from smallest (1) to largest (3) specific heat.

Metal	Initial Temperature (°C)	Final Temperature (°C)
A	20.0	58.6
B	20.0	37.2
C	20.0	32.3

1 smallest specific heat **A**
2 **B**
3 largest specific heat **C**

QUESTION 6 (5 points)

Which of the following equations shows how reactions **A – C** can be combined to generate reaction **D**?



A. ☒ **D = -A - 2B + 2C**

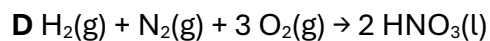
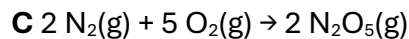
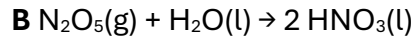
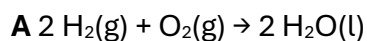
B. ☐ **D = A + 2B - 2C**

C. ☐ **D = A + B - 2C**

D. ☐ **D = -2A - B + 2C**

E. ☐ **D = A - B - C**

Which of the following equations shows how reactions **B – D** can be combined to generate reaction **A**?



A. ☒ **A = -2B - C + 2D**

B. ☐ **A = -2B + C + D**

C. ☐ **A = C - B + 2D**

D. ☐ **A = -2B + C - D**

E. ☐ **A = B - C - D**

QUESTION 7 (6 points)

Indicate the expected sign of the standard entropy ΔS° for each process below.

[☐ **Positive** ☐ Negative] $\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2 \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$

[☐ Positive ☐ **Negative**] $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightarrow 2 \text{NH}_3(\text{g})$

[☐ **Positive** ☐ Negative] $\text{CO}_2(\text{s}) \rightarrow \text{CO}_2(\text{g})$

Indicate the expected sign of the standard entropy ΔS° for each process below.

[☐ Positive ☐ **Negative**] $2 \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}_2(\text{aq})$

[☐ Positive ☐ **Negative**] $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$

[☐ Positive ☐ **Negative**] $2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{SO}_3(\text{g})$

Indicate the expected sign of the standard entropy ΔS° for each process below.

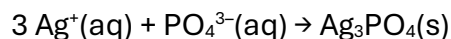
[☐ Positive ☐ **Negative**] $\text{CO}_2(\text{g}) \rightarrow \text{CO}_2(\text{s})$

[☐ Positive ☐ **Negative**] $2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{SO}_3(\text{g})$

[☐ **Positive** ☐ Negative] $\text{AgCl}(\text{s}) \rightarrow \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq})$

QUESTION 8 (6 points)

The process below occurs spontaneously from standard conditions. Indicate whether each statement below is true or false.

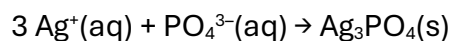


[☐ **True** ☐ False] For this process, ΔH_{sys}^0 is less than zero.

[☐ **True** ☐ False] The free energy of the product is less than the sum of the free energies of the reactants.

[☐ True ☐ **False**] The fact that the reaction is spontaneous implies that ΔS_{sys}^0 , the entropy change of the reaction system, is greater than zero.

The process below occurs spontaneously from standard conditions. Indicate whether each statement below is true or false.

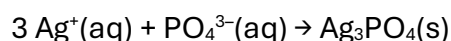


[☐ True ☐ **False**] For this process, ΔH_{sys}^0 is greater than zero.

[☐ **True** ☐ False] The free energy of the product is less than the sum of the free energies of the reactants.

[☐ **True** ☐ False] The fact that the reaction is spontaneous implies that ΔS_{univ}^0 , the total entropy change of the universe, is greater than zero.

The process below occurs spontaneously from standard conditions. Indicate whether each statement below is true or false.



[☐ True ☒ **False**] For this process, $\Delta H^\circ_{\text{sys}}$ is greater than zero.

[☐ True ☒ **False**] The free energy of the product is greater than the sum of the free energies of the reactants.

[☒ **True** ☐ False] The fact that the reaction is spontaneous implies that $\Delta S^\circ_{\text{univ}}$, the total entropy change of the universe, is greater than zero.

QUESTION 9 (5 points)

For which of the following combinations of signs of ΔH° and ΔS° is a reaction spontaneous in the forward direction from standard conditions only *below* a cutoff temperature?

- A. ☐ The temperature dependence of spontaneity cannot be determined from ΔH° and ΔS° alone.
 - B. ☐ $\Delta H^\circ > 0$, $\Delta S^\circ < 0$
 - C. ☒ $\Delta H^\circ < 0$, $\Delta S^\circ < 0$**
 - D. ☐ $\Delta H^\circ < 0$, $\Delta S^\circ > 0$
 - E. ☐ $\Delta H^\circ > 0$, $\Delta S^\circ > 0$
-

For which of the following combinations of signs of ΔH° and ΔS° is a reaction spontaneous in the forward direction from standard conditions only *above* a cutoff temperature?

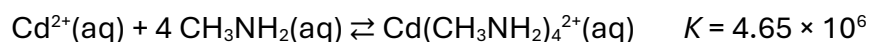
- A. ☐ The temperature dependence of spontaneity cannot be determined from ΔH° and ΔS° alone.
 - B. ☐ $\Delta H^\circ > 0$, $\Delta S^\circ < 0$
 - C. ☐ $\Delta H^\circ < 0$, $\Delta S^\circ < 0$
 - D. ☐ $\Delta H^\circ < 0$, $\Delta S^\circ > 0$
 - E. ☒ $\Delta H^\circ > 0$, $\Delta S^\circ > 0$**
-

For which of the following combinations of signs of ΔH° and ΔS° is a reaction spontaneous in the forward direction from standard conditions at *all* temperatures?

- A. ☐ The temperature dependence of spontaneity cannot be determined from ΔH° and ΔS° alone.
- B. ☐ $\Delta H^\circ > 0$, $\Delta S^\circ < 0$
- C. ☐ $\Delta H^\circ < 0$, $\Delta S^\circ < 0$
- D. ☒ $\Delta H^\circ < 0$, $\Delta S^\circ > 0$
- E. ☐ $\Delta H^\circ > 0$, $\Delta S^\circ > 0$

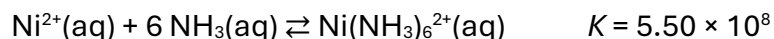
QUESTION 10 (5 points)

Consider the process below. What is ΔG° for this process at 293 Kelvin? Enter your response with appropriate sign in kilojoules per mole (kJ/mol) to the nearest 0.1 kJ/mol. The ideal gas constant R is 8.314 J/K·mol.



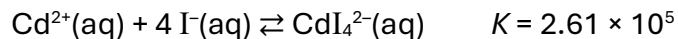
−37.4 kJ/mol

Consider the process below. What is ΔG° for this process at 293 Kelvin? Enter your response with appropriate sign in kilojoules per mole (kJ/mol) to the nearest 0.1 kJ/mol. The ideal gas constant R is 8.314 J/K·mol.



−49.0 kJ/mol

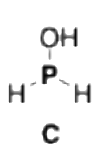
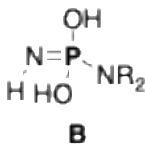
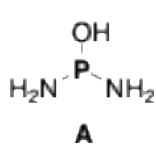
Consider the process below. What is ΔG° for this process at 293 Kelvin? Enter your response with appropriate sign in kilojoules per mole (kJ/mol) to the nearest 0.1 kJ/mol. The ideal gas constant R is 8.314 J/K·mol.



−30.4 kJ/mol

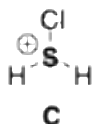
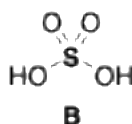
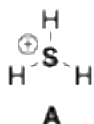
QUESTION 11 (6 points)

Rank the compounds below from most negative to most positive oxidation number at the bolded (phosphorus) atom. "R" is a generic carbon group (that is, carbon is linked to the indicated atom).



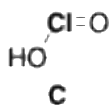
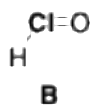
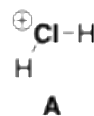
- 1 most negative oxidation number **C**
2 **A**
3 most positive oxidation number **B**
-

Rank the compounds below from most negative to most positive oxidation number at the bolded (sulfur) atom. "R" is a generic carbon group (that is, carbon is linked to the indicated atom).



- 1 most negative oxidation number **A**
2 **C**
3 most positive oxidation number **B**
-

Rank the compounds below from most negative to most positive oxidation number at the bolded (chlorine) atom. "R" is a generic carbon group (that is, carbon is linked to the indicated atom).



- 1 most negative oxidation number **A**
2 **B**
3 most positive oxidation number **C**

QUESTION 12 (6 points)

Consider the two half-reactions shown below. Indicate whether each is an oxidation or reduction process as written (electrons are omitted).

[☐ oxidation ☒ **reduction**] $\text{S}_2\text{O}_6^{2-} + 4 \text{H}^+ \rightarrow 2 \text{H}_2\text{SO}_3$

[☒ **oxidation** ☐ reduction] $\text{Ba} \rightarrow \text{Ba}^{2+}$

When combining the half-reactions above to create a balanced redox reaction with the smallest whole-number coefficients, how many electrons are transferred in the reaction?

[☐ 1 ☒ **2** ☐ 3 ☐ 4 ☐ 5 ☐ 6]

Consider the two half-reactions shown below. Indicate whether each is an oxidation or reduction process as written (electrons are omitted).

[☒ **oxidation** ☐ reduction] $2 \text{H}_2\text{SO}_3 \rightarrow \text{S}_2\text{O}_6^{2-} + 4 \text{H}^+$

[☒ **oxidation** ☐ reduction] $\text{Al} \rightarrow \text{Al}^{3+}$

When combining the half-reactions above to create a balanced redox reaction with the smallest whole-number coefficients, how many electrons are transferred in the reaction?

[☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ **6**]

Consider the two half-reactions shown below. Indicate whether each is an oxidation or reduction process as written (electrons are omitted).

[☐ oxidation ☒ **reduction**] $\text{SO}_2(\text{aq}) + 4 \text{H}^+ \rightarrow \text{S}(\text{s}) + 2 \text{H}_2\text{O}$

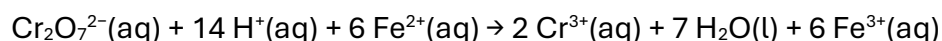
[☐ oxidation ☒ **reduction**] $\text{CH}_3\text{OH}(\text{aq}) + 2 \text{H}^+ \rightarrow \text{CH}_4(\text{g}) + \text{H}_2\text{O}$

When combining the half-reactions above to create a balanced redox reaction with the smallest whole-number coefficients, how many electrons are transferred in the reaction?

[☐ 1 ☐ 2 ☐ 3 ☒ **4** ☐ 5 ☐ 6]

QUESTION 13 (5 points)

Identify the oxidizing agent and reducing agent in the balanced redox reaction below.



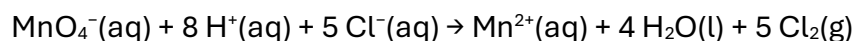
$\text{Cr}_2\text{O}_7^{2-}$ is the [☐ oxidizing agent ☐ reducing agent].

Fe^{2+} is the [☐ oxidizing agent ☐ reducing agent].

In this reaction, the protons (H^+) undergo

[☐ oxidation ☐ reduction ☐ no change in oxidation state].

Identify the oxidizing agent and reducing agent in the balanced redox reaction below.



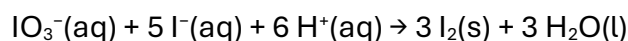
Cl^- is the [☐ oxidizing agent ☐ reducing agent].

MnO_4^- is the [☐ oxidizing agent ☐ reducing agent].

In this reaction, the protons (H^+) undergo

[☐ oxidation ☐ reduction ☐ no change in oxidation state].

Identify the oxidizing agent and reducing agent in the balanced redox reaction below.



I^- is the [☐ oxidizing agent ☐ reducing agent].

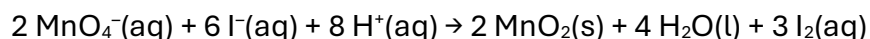
IO_3^- is the [☐ oxidizing agent ☐ reducing agent].

In this reaction, the protons (H^+) undergo

[☐ oxidation ☐ reduction ☐ no change in oxidation state].

QUESTION 14 (5 points)

For the reaction below, E° is +1.15 V. Fill in the blanks below.

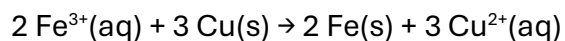


As written and starting from standard conditions, the reaction is
[☐ **spontaneous** ☐ non-spontaneous].

ΔG° for the reaction is [☐ positive ☒ **negative** ☐ zero].

K for the reaction is
[☒ **greater than 1** ☐ less than 1 ☐ equal to 1].

For the reaction below, E° is -0.38 V. Fill in the blanks below.



As written and starting from standard conditions, the reaction is
[☐ spontaneous ☒ **non-spontaneous**].

ΔG° for the reaction is [☒ **positive** ☐ negative ☐ zero].

K for the reaction is
[☐ greater than 1 ☒ **less than 1** ☐ equal to 1].

QUESTION 15 (5 points)

Which of the following metals will dissolve spontaneously in 1.0 M nitric acid (HNO_3)? In other words, for which of the metals is the reaction below spontaneous from standard conditions? Use the standard reduction potentials provided. Select all that apply.

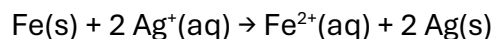


Half-reaction	$E^\circ_{\text{red}} (\text{V})$
$\text{Li}^+ + \text{e}^- \rightarrow \text{Li(s)}$	-3.04
$\text{Ba}^{2+} + 2 \text{e}^- \rightarrow \text{Ba(s)}$	-2.91
$\text{Pd}^{2+} + 2 \text{e}^- \rightarrow \text{Pd(s)}$	+0.915
$\text{Cs}^+ + \text{e}^- \rightarrow \text{Cs(s)}$	-3.026
$\text{Re}^{3+} + 3 \text{e}^- \rightarrow \text{Re(s)}$	+0.500

- A. ☐ Li
- B. ☐ Ba
- C. ☐ Pd
- D. ☐ Cs
- E. ☐ Re

QUESTION 16 (5 points)

A voltaic cell is constructed using iron metal, 1.00 M $\text{Fe}(\text{NO}_3)_2$, silver metal, and 1.00 M AgNO_3 . The salt bridge contains $\text{KNO}_3(\text{aq})$. The spontaneous redox reaction that occurs in the cell is below. Fill in the blanks to describe this cell.



Fe(s) and $\text{Fe}^{2+}(\text{aq})$ are found in the [☐ cathode ☒ **anode**].

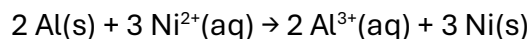
Ag(s) and $\text{Ag}^+(\text{aq})$ are found in the [☒ **cathode** ☐ anode].

Electrons flow from [☒ **Fe** ☐ Fe^{2+} ☐ Ag ☐ Ag^+]

to [☐ Fe ☐ Fe^{2+} ☐ Ag ☒ **Ag^+**] as the cell discharges.

As the cell discharges, K^+ ions migrate toward the [☒ **cathode** ☐ anode] and NO_3^- ions migrate toward the [☐ cathode ☒ **anode**].

A voltaic cell is constructed using aluminum metal, 1.00 M $\text{Al}(\text{NO}_3)_3$, nickel metal, and 1.00 M $\text{Ni}(\text{NO}_3)_2$. The salt bridge contains $\text{KNO}_3(\text{aq})$. The spontaneous redox reaction that occurs in the cell is below. Fill in the blanks to describe this cell.



$\text{Al}(\text{s})$ and $\text{Al}^{3+}(\text{aq})$ are found in the [☐ cathode ☒ **anode**].

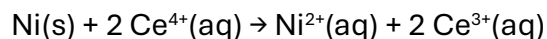
$\text{Ni}(\text{s})$ and $\text{Ni}^{2+}(\text{aq})$ are found in the [☒ **cathode** ☐ anode].

Electrons flow from [☒ **Al** ☐ Ni^{2+} ☐ Al^{3+} ☐ Ni]

to [☐ Al ☒ **Ni^{2+}** ☐ Al^{3+} ☐ Ni] as the cell discharges.

As the cell discharges, K^+ ions migrate toward the [☒ **cathode** ☐ anode] and NO_3^- ions migrate toward the [☐ cathode ☒ **anode**].

A voltaic cell is constructed using nickel metal, 1.00 M $\text{Ni}(\text{NO}_3)_2$, 1.00 M $\text{Ce}(\text{NO}_3)_4$, and 1.00 M $\text{Ce}(\text{NO}_3)_3$. The salt bridge contains $\text{KNO}_3(\text{aq})$. The spontaneous redox reaction that occurs in the cell is below. Fill in the blanks to describe this cell.



$\text{Ni}(\text{s})$ and $\text{Ni}^{2+}(\text{aq})$ are found in the [☐ cathode ☒ **anode**].

$\text{Ce}^{3+}(\text{aq})$ and $\text{Ce}^{4+}(\text{aq})$ are found in the [☒ **cathode** ☐ anode].

Electrons flow from [☒ **Ni** ☐ Ce^{4+} ☐ Ni^{2+} ☐ Ce^{3+}]

to [☐ Ni ☒ **Ce^{4+}** ☐ Ni^{2+} ☐ Ce^{3+}] as the cell discharges.

As the cell discharges K^+ ions migrate toward the [☒ **cathode** ☐ anode] and NO_3^- ions migrate toward the [☐ cathode ☒ **anode**].

QUESTION 17 (5 points)

Which of the following equations involving a thermodynamic state or path function is *NOT* correct?

A. ☐ $H = U + PV$

B. ☐ $w = -P \Delta V$

C. ☐ $q = C \Delta T$

D. ☐ $G = H - TS$

E. ☒ $\Delta G = -E/nF$

Which of the following equations involving a thermodynamic state or path function is *NOT* correct?

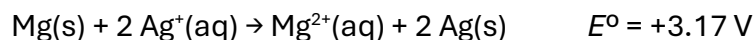
- A. ☐ $H = U - PV$
 - B. ☐ $w = -P \Delta V$
 - C. ☐ $q = C \Delta T$
 - D. ☐ $G = H - TS$
 - E. ☐ $\Delta G = -nFE$
-

Which of the following equations involving a thermodynamic state or path function is *NOT* correct?

- A. ☐ $H = U + PV$
- B. ☐ $w = P \Delta V$
- C. ☐ $q = C \Delta T$
- D. ☐ $G = H - TS$
- E. ☐ $\Delta G = -nFE$

QUESTION 18 (5 points)

Consider a galvanic cell based on the following spontaneous redox reaction under standard conditions. Indicate whether each intervention (on the standard cell, changing nothing else) will cause an increase, decrease, or no change in the cell potential.

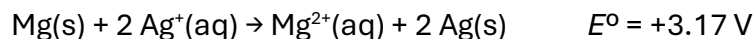


[☐ **increase** ☐ decrease ☐ no change] Increasing the molarity of Ag^+

[☐ increase ☐ **decrease** ☐ no change] Increasing the molarity of Mg^{2+}

[☐ increase ☐ decrease ☐ **no change**] Increasing the temperature

Consider a galvanic cell based on the following spontaneous redox reaction under standard conditions. Indicate whether each intervention (on the standard cell, changing nothing else) will cause an increase, decrease, or no change in the cell potential.



[☐ increase ☐ decrease ☒ **no change**] Increasing the mass of magnesium metal

[☒ **increase** ☐ decrease ☐ no change] Decreasing the molarity of Mg^{2+}

[☐ increase ☐ decrease ☒ **no change**] Decreasing the temperature

QUESTION 19 (5 points)

A current of 0.100 A was applied for 1.00 hour for the electrolytic oxidation of an unknown metal to aqueous metal cation. During this process, 1.87×10^{-3} mol of the metal cation were produced. What is the charge on the metal cation? Faraday's constant is 96485 C/mol.

- A. ☐ +1
- B. ☒ **+2**
- C. ☐ +3
- D. ☐ +4
- E. ☐ +5

A current of 0.100 A was applied for 1.00 hour for the electrolytic oxidation of an unknown metal to aqueous metal cation. During this process, 3.73×10^{-3} mol of the metal cation were produced. What is the charge on the metal cation? Faraday's constant is 96485 C/mol.

- A. ☒ **+1**
 - B. ☐ +2
 - C. ☐ +3
 - D. ☐ +4
 - E. ☐ +5
-

A current of 0.100 A was applied for 1.00 hour for the electrolytic oxidation of an unknown metal to aqueous metal cation. During this process, 1.24×10^{-3} mol of the metal cation were produced. What is the charge on the metal cation? Faraday's constant is 96485 C/mol.

- A. ☐ +1
- B. ☐ +2
- C. ☒ +3**
- D. ☐ +4
- E. ☐ +5